

Conductive Rigid Skeleton Supported Silicon as High-Performance Li-Ion Battery Anodes

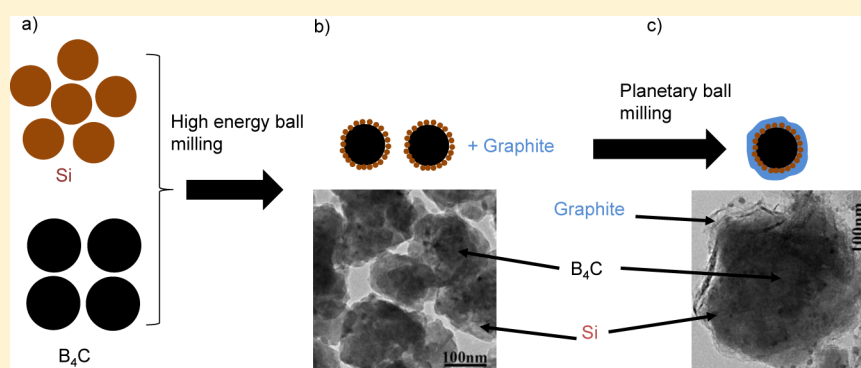
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S Supporting Information



ABSTRACT: A cost-effective and scalable method is developed to prepare a core-shell structured Si/B₄C composite with graphite coating with high efficiency, exceptional rate performance, and long-term stability. In this material, conductive B₄C with a high Mohs hardness serves not only as micro/nano-millers in the ball-milling process to break down micron-sized Si but also as the conductive rigid skeleton to support the in situ formed sub-10 nm Si particles to alleviate the volume expansion during charge/discharge. The Si/B₄C composite is coated with a few graphitic layers to further improve the conductivity and stability of the composite. The Si/B₄C/graphite (SBG) composite anode shows excellent cyclability with a specific capacity of ~822 mAh·g⁻¹ (based on the weight of the entire electrode, including binder and conductive carbon) and ~94% capacity retention over 100 cycles at 0.3 C rate. This new structure has the potential to provide adequate storage capacity and stability for practical applications and a good opportunity for large-scale manufacturing using commercially available materials and technologies.

KEYWORDS: Silicon, anode, rigid skeleton, core-shell structure, lithium-ion batteries, energy storage, boron carbide

Hybrid electric vehicles (HEVs) and plug-in hybrid electric vehicles (PHEVs) have great potential to reduce the environmental impact related to the use of fossil fuels. To meet the requirements set for electric vehicles, high-energy-density lithium (Li)-ion batteries need to be developed.^{1–4} Silicon (Si) is a promising anode material for the next generation Li-ion batteries because its Li-ion storage capacity (~4200 mAh·g⁻¹ for Li) is much higher than that of conventional graphite anodes used in commercial Li-ion batteries (372 mAh·g⁻¹).^{5,6} However, Si-based anodes suffer from structural failure and pulverization of the Si particles due to the large volume change (300% to 400%) during the lithiation and delithiation processes.⁷

In the past decade, great efforts have been made to reduce the Si particle size or produce three-dimensional (3D) porous electrode structures to avoid the pulverization and improve the stability of the Si-based anodes. Various nanostructures are synthesized, including Si nanowires⁸ and conductive-core/Si-shell core-shell structured nanowires^{9,10} fabricated by chemical

vapor deposition (CVD), 3D porous Si anodes prepared by chemical deposition of Si on templates,^{11–13} tobacco mosaic virus (TMV) templated 3D Si anodes,^{14–16} carbon¹⁷ or polymer¹⁸ scaffold-supported Si anodes, and Si-carbon nanocomposite granules formed through hierarchical bottom-up assembly.¹⁹ Although some properties of these Si-based anodes far exceed those of the conventional anode materials and demonstrated very high capacity and good stability (especially those of the nanostructured Si nanowires or hollow structures),^{20–22} overall performance (especially the electrode loading or the capacity per unit area) of these Si-based anode materials still cannot satisfy the needs for practical applications.¹⁰ Another significant challenge is to develop

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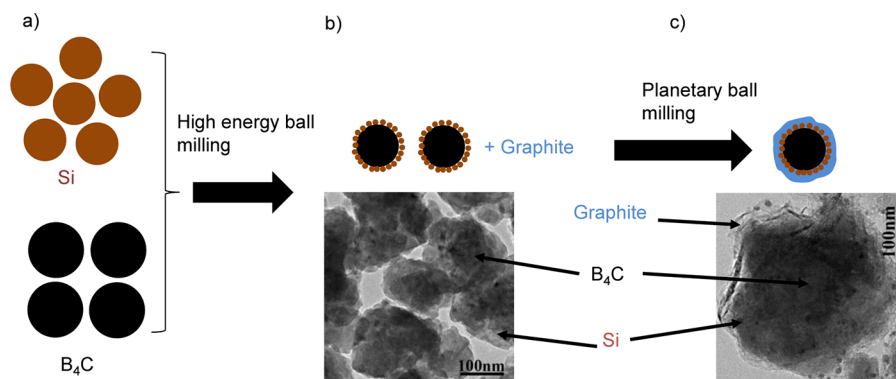


Figure 1. Schematic diagram of the synthesis process of conductive-rigid-skeleton-supported Si with TEM images for the intermediate product of Si/ B_4C and the final SBG product. (a) Starting materials of micron-sized B_4C and Si. (b) Schematic diagram of the Si/ B_4C core-shell structure and TEM image. (c) Schematic diagram of the SBG structure and TEM image.

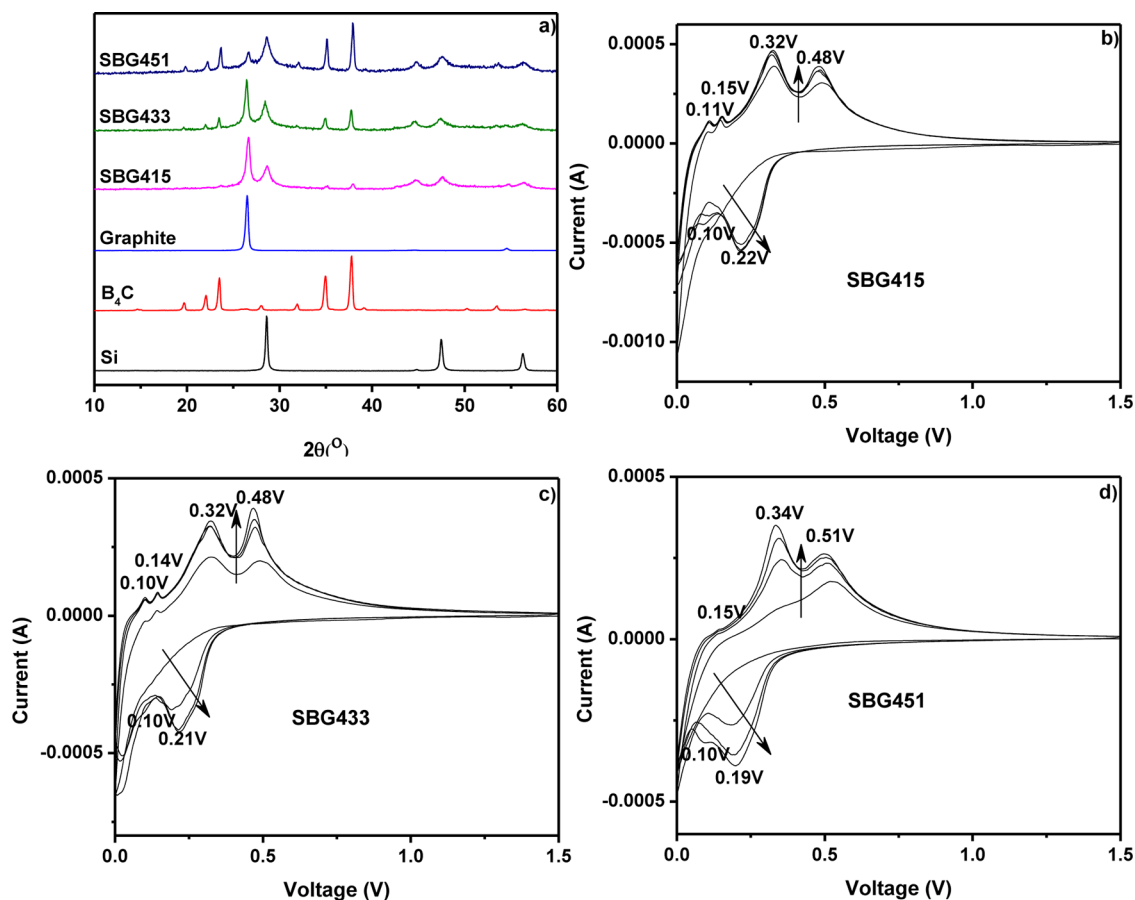


Figure 2. Characterization of the SBG with different compositions. (a) XRD patterns. (b) CV profiles of SBG415. (c) CV profiles of SBG433. (d) CV profiles of SBG451.

materials and methods that are cost-effective and applicable for large-scale manufacturing.

In this work, we developed a Si/ B_4C /graphite (SBG) core-shell-shell structured composite using simple ball-milling (BM) of commercially available materials and demonstrated that such materials have good cycling stability. Highly conductive boron carbide (B_4C) is used as nano/micro-millers to break down micron-sized Si and also as a conductive skeleton in the final composite to support the Si. The graphitic coating on the Si/ B_4C composite improves the electrical conductivity and favors formation of the solid electrolyte interphase (SEI) film.²³ The SBG composite anode exhibits

excellent long-term stability, rate performance, and Coulombic efficiency. The concept of using conductive hard skeleton-supported Si opens a new route for anodes with high capacity, long cycling stability, and excellent rate performance. The starting materials and the synthetic processes are both viable for large-scale production, making this approach particularly attractive for practical applications.

Figure 1 shows the schematic diagram of the synthesis process of SBG composite. Micron-sized Si and B_4C were mixed in an appropriate ratio and mechanically milled (Figure 1a) to form the Si/ B_4C composite. The transmission electron microscopy (TEM) image of the intermediate product of Si/

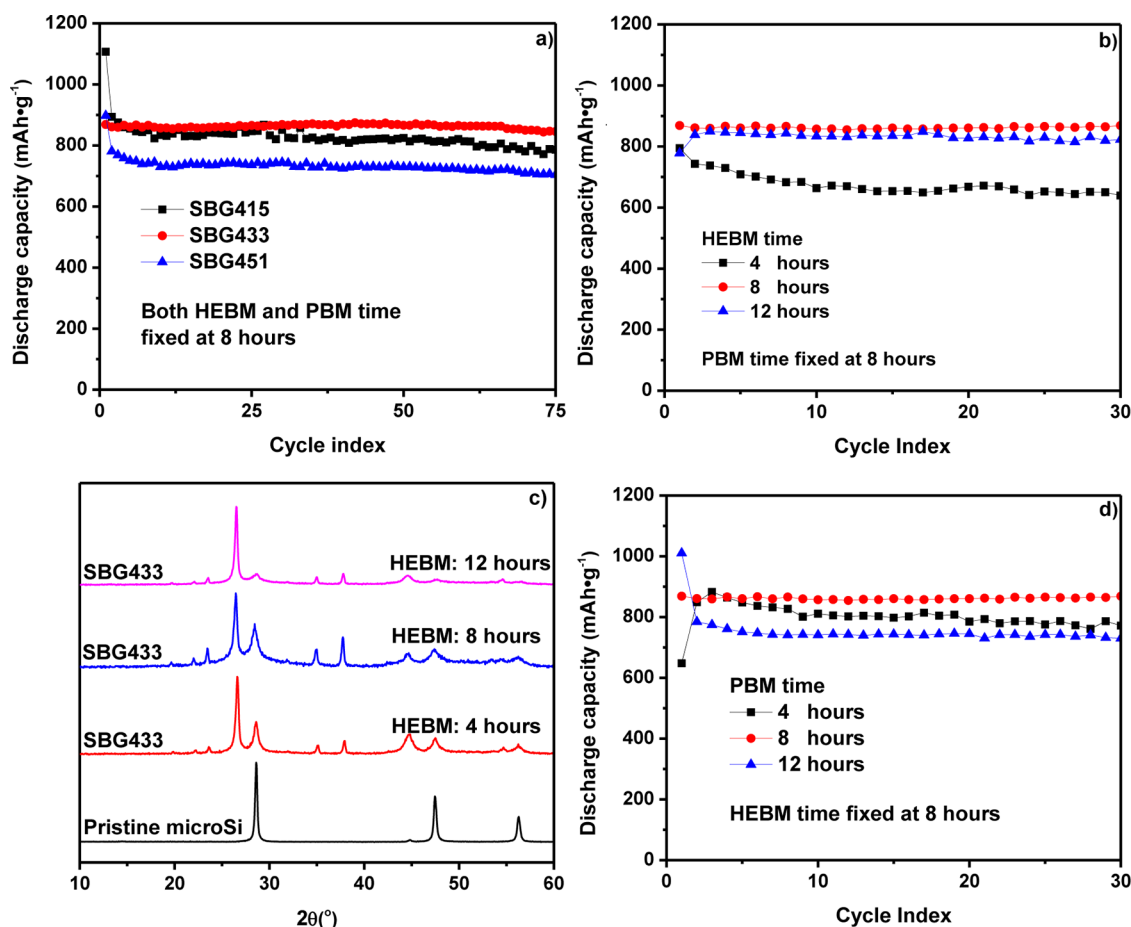


Figure 3. Effect of SBG composition and the synthesis condition on the cycling stability at a current density of $0.63 \text{ A} \cdot \text{g}^{-1}$ based on the silicon weight. (a) Comparison of the long-term cycling stability for SBG composites with different compositions; (b) cycling stability of SBG433 with different high energy ball milling (HEBM) times; (c) XRD patterns of the SBG433 with different HEBM times; (d) cycling stability of SBG433 with different PBM times.

B₄C shows the size of the Si particles has been significantly reduced from 1–5 μm to <10 nm and the particle size of the conductive skeleton material B₄C is reduced from 1–7 μm to 100–300 nm (Figure 1b). The in situ generated nanosized Si particles were attached to the B₄C particles, forming the Si-coated B₄C core–shell structure. Graphite coating on the Si/B₄C core–shell structure was achieved by a subsequent planetary ball milling (PBM) of Si/B₄C and graphite, in which graphite was mechanically peeled, forming a graphite coating on the surface of Si/B₄C. The TEM image in Figure 1c clearly shows that the core–shell structured Si/B₄C composite is covered by another shell, a thin layer of graphite, to form a three-layer core–shell–shell structure. The thin graphite coating contains only a few layers of graphene sheets as shown in Figure 1c and Figure S1 of the Supporting Information. This observation is consistent with a previous study on silicon-carbide-supported materials.²⁴

The crystalline structures of the precursors and SBG composites with different compositions were characterized by X-ray diffraction (XRD). B₄C, graphite, and Si peaks are indexed in the SBG composites. As shown in Figure 2a, the intensity of the graphite peaks decreases when the graphite content decreases from 50 wt % in SBG415, to 30 wt % in SBG433, to 10 wt % in SBG451, while the peak intensity of B₄C increases when the B₄C content increases from 10 wt % in SBG415, to 30 wt % in SBG433, and 50 wt % in SBG451. All of

these three composites have the same Si ratio, that is, 40 wt %. The broad Si peaks (Figure 2a) in the composites suggest that the Si particles were significantly reduced in size although they still maintain their crystalline structure after the ball-milling processes. This is consistent with the results of cyclic voltammetry (CV) analysis (Figures 2b–d) discussed below. There is no visible change in the characteristic peaks for B₄C particles even though their sizes have been decreased after ball milling, as observed in the TEM images (see Figure 1). This result can be attributed to the smaller size changes for B₄C particles (as compared with those of Si) because of their high Mohs hardness.

To understand the effect of different B₄C/graphite ratios on the electrochemical performance of SBG composites, these samples were systematically investigated by a CV scan (Figures 2b–d). All of these composites show similar electrochemical properties. B₄C is electrochemically inactive and did not contribute to the capacity under our experimental conditions (Figure S2 of the Supporting Information). In the first scan, one cathodic peak ($\sim 0 \text{ V}$) and four anodic peaks (at $\sim 0.11 \text{ V}$, $\sim 0.15 \text{ V}$, $0.32\text{--}0.34 \text{ V}$, and $0.48\text{--}0.51 \text{ V}$) are observed. In the subsequent scans, another two cathodic peaks at $\sim 0.10 \text{ V}$ and $\sim 0.20 \text{ V}$ appear. The cathodic peaks at $\sim 0.10 \text{ V}$ and the anodic peaks at $\sim 0.11 \text{ V}$ and $\sim 0.15 \text{ V}$ are from graphite in the composites.^{25,26} The cathodic peak at $\sim 0.20 \text{ V}$ and the anodic peaks at $0.32\text{--}0.34 \text{ V}$ and $0.48\text{--}0.51 \text{ V}$ are characteristic peaks

for amorphous Si, and the cathodic peak at ~ 0 V is for both crystalline and amorphous Si.^{8,27,28} In the first cathodic scan, only crystalline Si exists in the composites, so only one peak at ~ 0 V is observed. However, the crystalline structure converts to amorphous during Li insertion, and two anodic peaks (at 0.32–0.34 V and 0.48–0.51 V) characteristic of amorphous Si are observed in the subsequent anodic scans. The slight increase in the intensity of the peaks (as indicated by the arrows in the Figure 2b–d) can be attributed to the small amount of remaining crystalline Si when it is converted into amorphous phase during the repeated lithiation and delithiation. The cathodic peak for amorphous Si moves negatively from 0.22 V for SBG415, to 0.21 V for SBG433, and to 0.19 V for SBG451. At the same time, the anodic peaks for amorphous Si move positively from 0.32 and 0.48 V for SBG415 and SBG433 to 0.34 and 0.51 V for SBG451. This voltage shift may be due to an electronic resistance increase since the graphite content decreases from 50% in SBG415 to 10% in SBG451 and this resistance increase increases the polarization potential. The intensity of the anodic peaks for graphite also decreases with the decrease of the graphite component and is barely visible for SBG451. The cathodic peak at ~ 0.10 V for graphite is not visible for some scans because it is overlapped by the cathodic peak at ~ 0 V for Si.

The thicknesses of both the Si coating and the graphite coating on B_4C can affect the cycling stability. Figure 3a shows that all of these three samples show good stability and high capacities around $800 \text{ mAh}\cdot\text{g}^{-1}$ based on the weight of the entire electrode including binder and conductive carbon. After 75 cycles, the capacity retention is $\sim 88.0\%$ for SBG415, 98.3% for SBG433, and 90.0% for SBG451. The capacity retention is 63.9% after 50 cycles for the control sample SB46 (40% silicon and 60% B_4C , Figure S3 of the Supporting Information). Since there is irreversible capacity in the first cycle due to the formation of the SEI layer, the discharge capacity in the second cycle is used for the capacity retention calculation. The fact that SBG433 shows the best stability and the highest capacity can be correlated to the optimized ratio between the Si, B_4C , and graphite components.

As described in the experimental section, the B_4C component in the composites increases in the order of SBG415 < SBG433 < SBG451, while the graphite component decreases to maintain the same Si ratio in composites. The increased amount of B_4C in the samples will provide more surface area for Si nanoparticles to attach to. Since the amount of Si remains the same in all of these composites, the thickness of the Si layer on B_4C particles will decrease in the order of SBG415 > SBG433 > SBG451. SBG composites with thinner Si layers would experience smaller volume change during lithiation and delithiation and be expected to give more stable cycling.⁷ In other words, the cycling stability of SBG433 will be better than that of SBG415. However, a further increase in the amount of B_4C will lead to a large decrease in the amount of graphite, which can effectively alleviate the stress generated in the lithiation and delithiation and thus help to stabilize the integrity of the electrode.^{17,29–34} A Si/ B_4C /graphite ratio of 4:3:3 (sample SBG433) provides a good balance between the thickness of the Si coating and the thickness of the conductive graphite coating; therefore sample SBG433 demonstrates a much better stability than other samples investigated in this work (Figure 3a).

Another interesting phenomenon observed in this work is that the first-cycle Coulombic efficiency of the samples

increases with decreasing amounts of graphite. The sample SBG415, which contains 50% graphite, exhibits a first-cycle efficiency of 78.1%. When the graphite weight ratio decreases to 30% (SBG433) and 10% (SBG451), the corresponding first-cycle efficiency increases to 82.3% and 84.6%, respectively. This is because the decreased graphite ratio leads to a reduced surface area, which in turn leads to decreased side reaction and therefore a higher irreversible capacity.³⁵ This explanation is consistent with the Brunauer–Emmett–Teller (BET) measurement results which show that the composites' surface areas decrease in the order of SBG415 ($151.8 \text{ m}^2\cdot\text{g}^{-1}$) > SBG433 ($88.2 \text{ m}^2\cdot\text{g}^{-1}$) > SBG451 ($44.5 \text{ m}^2\cdot\text{g}^{-1}$). As a result, the SBG core–shell–shell structure with an optimized composition ratio greatly enhances the cycling stability and first-cycle Coulombic efficiency.

Ball-milling time is another key factor that has significant impact on the stability of the SBG composite. Figure 3b–d shows the effects of different ball-milling times on the stability of SBG433. First, HEBM was carried out for 4, 8, and 12 h while the PBM time was fixed at 8 h. As shown in Figure 3b, the sample using 4-h HEBM shows relatively worse stability than other two samples using 8-h and 12-h ball-milling. Its capacity retention after 30 cycles is 86.1% compared to near 100% for the samples using 8-h and 12-h HEBM. Therefore, the HEBM time needs to be long enough to break down the micron-sized Si particles to nano size so they can tolerate the large volume changes during the lithiation and delithiation processes.³⁶ As shown in Figure 3c, the widths of the characteristic peaks for Si become broader when HEBM time is increased. The broader Si peaks indicate smaller Si particles according to the Scherrer equation. The samples using 8-h and 12-h HEBM have a very similar capacity retention at $\sim 100\%$. This means that an 8-h HEBM time is enough to obtain the desired Si particle size with appropriate Si/ B_4C structure. Figure 3d shows the results obtained while the HEBM time is fixed at 8 h and the PBM time is changed from 4 to 8 h and 12 h. The best capacity and capacity retention after 30 cycles is obtained when PBM time is controlled at 8 h. A short PBM time leads to slightly lower stability because it is not enough to establish sufficient graphite coverage on the Si/ B_4C particles. On the other hand, too long a PBM time may remove some nano-Si particles from the B_4C surface and lead to a decreased specific capacity as shown in Figure 3d. The effect of graphite on the conductivity of the SBG electrode is demonstrated by the dramatic change in the impedance of Si/ B_4C composite before and after graphite coating. As shown in Figure 4, the first semicircle at the high-to-medium frequency range corresponds to the impedance of the SEI film; the second semicircle at the medium-to-low frequency range corresponds to the impedance of the charge transfer occurring at the surface of the electrode; and the inclined line at the low frequency range corresponds to the lithium ion diffusion in the bulk electrode.³⁷ With graphite coating, both impedance of the charge transfer and the SEI film have been significantly reduced, which corroborates that the graphite coating plays here two important roles: the first one is the graphite coating can enhance the electronic conductivity of the electrode and thus make the surface charge transfer more easily;^{17,37} the second one is that the graphite coating can help form a stable and dense SEI film and thus the impedance caused by the SEI film is reduced.¹⁷

Based on the above discussion, a good component ratio and reasonable milling time are important to produce Si nanoparticles that are small enough to withstand the volume change

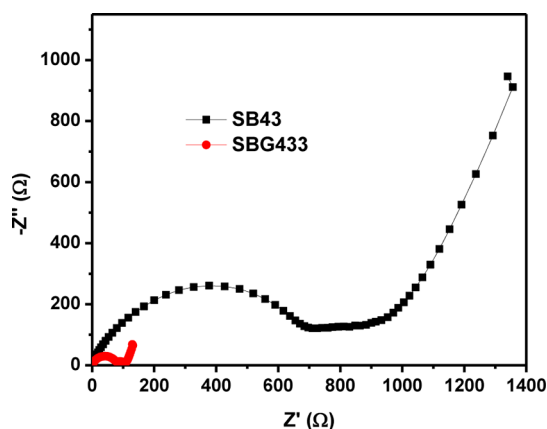


Figure 4. Nyquist plots of cells after the first lithiation at a current density of $3.75 \text{ A}\cdot\text{g}^{-1}$, using a mixture of Si and B_4C (weight ratio 4:3, labeled as SB43) or a mixture of Si, B_4C , and graphite (weight ratio 4:3:3, labeled as SBG433) as active materials. The SBG433 shows significantly lower total impedance than SB43 because of graphite coating.

at the micro level, attach these particles to a hard skeleton structure, and form a layer of good graphite on Si/ B_4C to provide a highly conductive coating and tolerate the volume change at the macro level due to its cushion effect. SBG composites with optimized ratio have a high capacity and very good long-term stability. Figure 5 shows the discharge–charge profiles, long-term stability, and rate performance of SBG433 prepared by 8-h HEBM and 8-h PBM. As shown in Figure 5a, the discharge capacity based on the entire electrode weight is $868.8 \text{ mAh}\cdot\text{g}^{-1}$ at the first cycle and $815.5 \text{ mAh}\cdot\text{g}^{-1}$ at the 100th cycle. This is equivalent to a specific capacity of $3102 \text{ mAh}\cdot\text{g}^{-1}$ (initial) and $2912 \text{ mAh}\cdot\text{g}^{-1}$ (at the 100th cycle) based on Si weight. The discharge capacity loss over 100 cycles is only $\sim 0.06\%$ per cycle. The capacity retention of SBG433 after 200 cycles is 78.5% . The Coulombic efficiency increases from $\sim 82.3\%$ at the first cycle to $\sim 97.8\%$ at the third cycle, 99.0% at tenth cycle, and then stays above 99.0% thereafter (Figure 5b). Due to the good electrical conductivity of the B_4C skeleton ($>140 \text{ S/m}$) and the graphite coating (see Supporting Information), SBG433 has exceptional rate performance (Figure 5c). The average remaining capacity based on the entire electrode weight including binder and conductive carbon is $900.1 \text{ mAh}\cdot\text{g}^{-1}$ at $0.31 \text{ A}\cdot\text{g}^{-1}$, $822.5 \text{ mAh}\cdot\text{g}^{-1}$ at $0.63 \text{ A}\cdot\text{g}^{-1}$, $723.6 \text{ mAh}\cdot\text{g}^{-1}$ at $1.25 \text{ A}\cdot\text{g}^{-1}$, and $601.2 \text{ mAh}\cdot\text{g}^{-1}$ at $2.50 \text{ A}\cdot\text{g}^{-1}$. The current densities are based on the weight of the Si component. The average remaining capacity based on Si weight is $3215 \text{ mAh}\cdot\text{g}^{-1}$ at $0.31 \text{ A}\cdot\text{g}^{-1}$, $2938 \text{ mAh}\cdot\text{g}^{-1}$ at $0.63 \text{ A}\cdot\text{g}^{-1}$, $2584 \text{ mAh}\cdot\text{g}^{-1}$ at $1.25 \text{ A}\cdot\text{g}^{-1}$, and $2147 \text{ mAh}\cdot\text{g}^{-1}$ at $2.5 \text{ A}\cdot\text{g}^{-1}$. When the current density is changed from $2.50 \text{ A}\cdot\text{g}^{-1}$ back to $0.31 \text{ A}\cdot\text{g}^{-1}$, the discharge capacity is recovered; this excellent capacity recovery further verifies the excellent rate performance of SBG composite with a core–shell–shell structure.

It should be noted that it is possible to further increase the specific capacity of the SBG anode by optimizing the particle sizes of both Si and B_4C and the content of the active material in the nanocomposites. However, for practical applications, the specific capacity of the cathode materials is still quite limited. A stable specific capacity of $800 \text{ mAh}\cdot\text{g}^{-1}$ based on the entire anode materials might be sufficient for the cathode chemistry that is currently accessible.

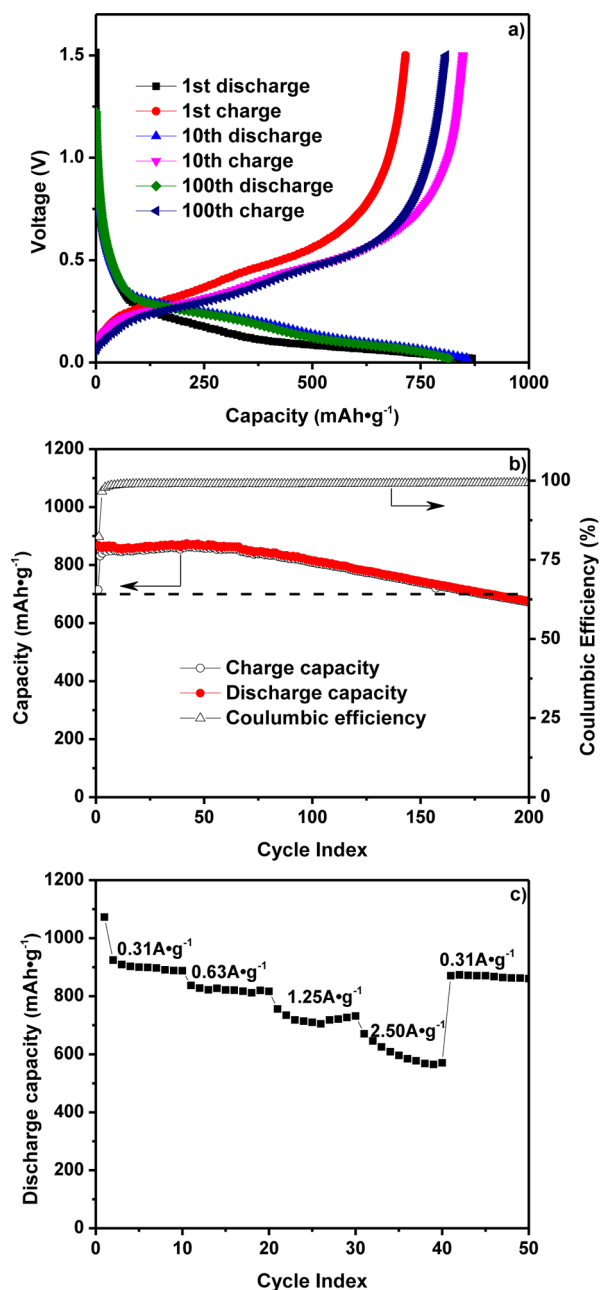


Figure 5. Electrochemical performance of SBG433 with optimized composition and ball-milling conditions: (a) discharge–charge profiles, (b) long-term cycling stability of SBG433 at a current density of $0.63 \text{ A}\cdot\text{g}^{-1}$, and (c) rate performance for SBG433 prepared by 8-h HEBM and 8-h PBM milling time.

In summary, a conductive-rigid-skeleton-supported Si composite was developed as an anode for Li-ion battery applications. The core–shell–shell structured SBG composite demonstrates high capacity and excellent stability. It exhibits only 6% capacity loss over 100 cycles. The specific capacity of the SBG composite is $822.5 \text{ mAh}\cdot\text{g}^{-1}$ at $0.63 \text{ A}\cdot\text{g}^{-1}$ and $601.2 \text{ mAh}\cdot\text{g}^{-1}$ at $2.50 \text{ A}\cdot\text{g}^{-1}$ based on the weight of the entire electrode including the weight of binder and conductive carbon. The specific capacity based on Si is $2938 \text{ mAh}\cdot\text{g}^{-1}$ at $0.63 \text{ A}\cdot\text{g}^{-1}$ and $2147 \text{ mAh}\cdot\text{g}^{-1}$ at $2.50 \text{ A}\cdot\text{g}^{-1}$. The SBG composite with an optimized graphite shell exhibits an initial Coulombic efficiency of 82.3% . The approach of using conductive hard material as nanomillers to in situ form sub- 10 nm Si particles and

subsequently form a conductive-rigid-skeleton-supported Si-based composite can be a general method for fabricating high performance electrodes for Li-ion batteries. Both the source materials and the preparation approaches used in this work are cost-effective and easy to scale up. Therefore, it has a good potential to be used for large scale applications. Furthermore, the approach reported in this work can also be used to stabilize other functional nanocomposite materials which may experience large volume expansion during physical, chemical, or electrochemical operations.

Methods and Materials. Micron-sized Si (1–5 μm , Sigma-Aldrich), micron-sized B_4C (1–7 μm , Alfa Aesar), and micron-sized graphite (reagent grade, Sigma-Aldrich) were used as received. As shown in Figure 1, the SBG composite was prepared by ball milling the mixture of Si and B_4C powders in a high energy ball-mill (8000 M Mixer/Mill, SPEX, US) and then by ball milling the Si/ B_4C composite with graphite in a planetary ball mill (Retsch, PM 100 CM) at 400 rpm.

To investigate the effect of the compositions on the anode performance, the weight ratio of Si: B_4C :graphite in the composite was varied systematically, including 4:1:5 (labeled as SBG415), 4:3:3 (labeled as SBG433), and 4:5:1 (labeled as SBG451). The different milling times (4 h, 8 h, and 12 h) for both HEBM and PBM were used to investigate the effect of milling time on the performance of the samples.

The as-prepared composites were characterized by XRD (Philips X'Pert X-ray diffractometer), TEM (JEOL-2010), and BET (Quantachrome Autosorb 6-B). The electrode sheet was prepared by casting a slurry of SBG composite, super P (SP) (Timcal), and carboxymethyl cellulose sodium salt (Na-CMC) (Kynar HSV900, from Arkema Inc.) solution (2.5 wt %) in distilled water onto copper foil (All Foils, Inc.). The weight ratio of SBG, SP, and Na-CMC was 7:1:2. After the solvent (water) was evaporated, the electrode sheet was die-cut into disks with a diameter of 1.27 cm and dried overnight under vacuum at 110 $^{\circ}\text{C}$.

Half cells were assembled in an argon-filled MBraun glovebox using Li metal as the counter electrode, Celgard K1640 as a separator, and 1 M LiPF_6 in ethylene carbonate and dimethyl carbonate (1:2 ratio in volume) with 10 wt % fluoroethylene carbonate additive as the electrolyte. The electrochemical performance of the coin cells was measured at room temperature using a battery tester (model BT-2000, Arbin Instruments). The cells were cycled between 0.02 and 1.5 V. Cyclic voltammetry scans were conducted on a CHI 1000A impedance analyzer (CH Instruments) at a scan rate of 0.05 $\text{mV}\cdot\text{s}^{-1}$ in a voltage range of 0–1.5 V using a two-electrode configuration. Fresh cells were used in the CV study, and four continuous full scans were conducted. The charge/discharge current density is calculated based on the weight of Si in the electrode.

■ ASSOCIATED CONTENT

■ Supporting Information

TEM image of the composite, capacity of the control electrodes, and cycling stability of the control composite. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

J.Z., J.L., X.C., Y.C., and X.L. conceived and designed the experiments. X.C. did the materials synthesis, electrode preparation, cell assembly, and characterization. X. L, W.X., and F.D. prepared the electrolyte. X.L. did the microscopic and XRD work. X.C. wrote the manuscript. X.L, J.Z., J.L., G.L.G., J.X., and P.M. edited the manuscript. This work was supported by the Assistant Secretary for Energy Efficiency and Renewable Energy, Office of Vehicle Technologies of the U.S. Department of Energy under Contract No. DE-AC02-05CH11231, Subcontract No. 18769 under the Batteries for Advanced Transportation Technologies (BATT) program. A portion of the research was performed in EMSL, a national scientific user facility sponsored by the Department of Energy's Office of Biological and Environmental Research and located at Pacific Northwest National Laboratory.

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